



Global Climate Exploratory Data Analysis

Analyzing Temperature Trends Using Python



Project Overview

This project explores long-term global climate trends using exploratory data analysis techniques in Python.

- Data Cleaning
- Data Formatting
- Trend Analysis
- Distribution Analysis
- Outlier Detection
- Climate Uncertainty Visualization



Dataset Information

The dataset contains historical land and ocean temperature measurements across multiple years.

Column	Description
dt	Date Column
LandAndOceanAverageTemperature	Global Average Temperature
LandAndOceanAverageTemperatureUncertainty	Confidence Interval Around Temperature



Data Cleaning

Several preprocessing techniques were applied before analysis.

- Handled missing values

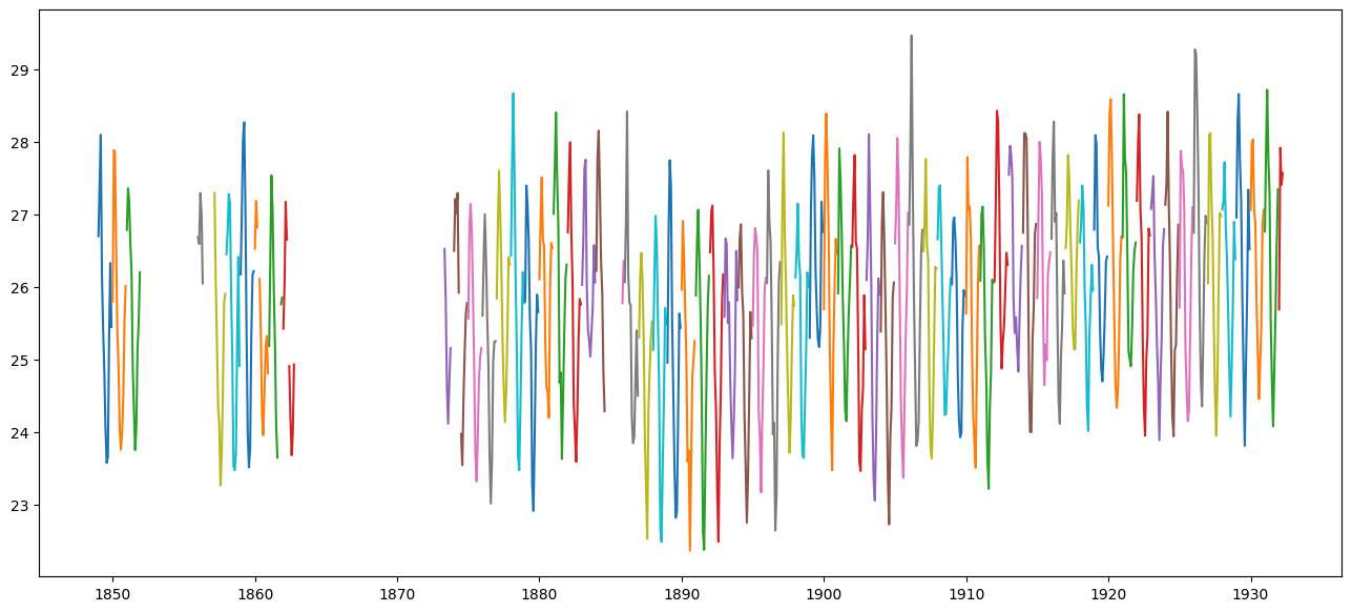
- Converted date columns
- Formatted datetime objects
- Explored data types
- Prepared dataset for visualization



Global Temperature Trend

The line graph below visualizes long-term global temperature patterns over time.

- Gradual warming trend observed
- Temperature fluctuations visible
- Recent decades appear warmer

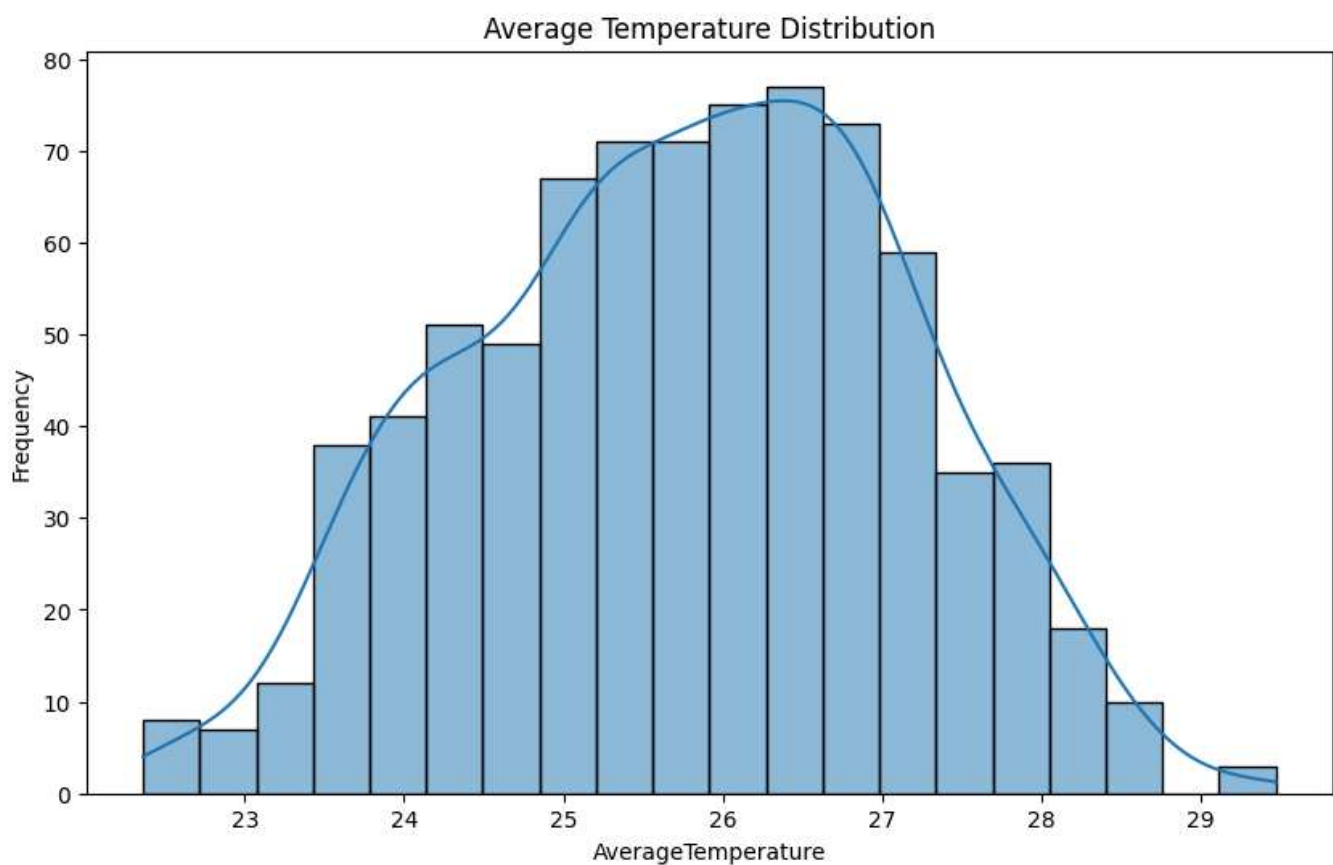




Temperature Distribution

Histogram analysis was used to understand how temperatures are distributed across the dataset.

- Most temperatures cluster near central values
- Moderate spread observed
- No severe skewness detected

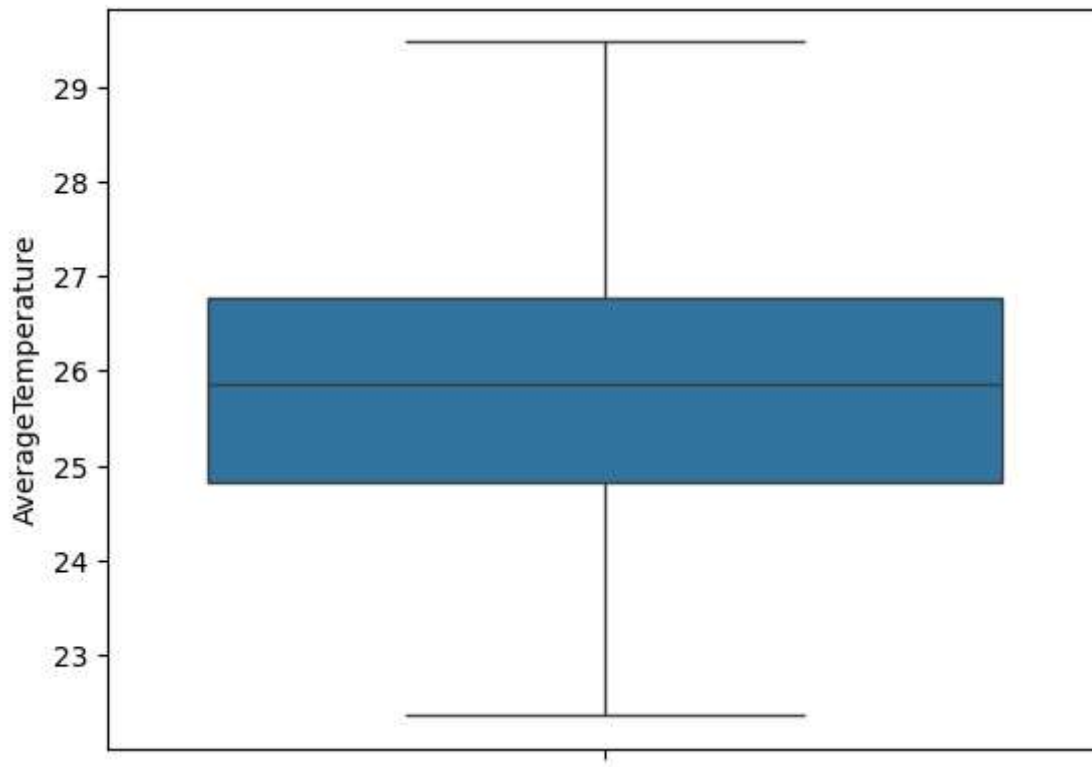


Boxplot Analysis

Boxplots were used to analyze spread, quartiles, and potential outliers.

- Median temperature identified
- IQR visualized using box region
- Extreme observations detected as outliers

<Axes: ylabel='AverageTemperature'>



Skills Demonstrated

- Exploratory Data Analysis
- Pandas Data Manipulation
- Matplotlib Visualization
- Seaborn Visualization
- Datetime Handling
- Statistical Interpretation

Conclusion

This project explored long-term global climate behavior using exploratory data analysis and visualization techniques in Python.

The analysis revealed warming patterns, temperature distribution behavior, and uncertainty trends across time.

★ Thank You For Viewing This Project ★

Feel free to explore the notebook and visualizations.